Workshop I

Introduction to Soldering

© 2025 Open Project Space, Institute of Electrical and Electronics Engineers at the University of California, Irvine. All Rights Reserved.

SECTION I

What is Soldering?

Nasa Video (20 mins)



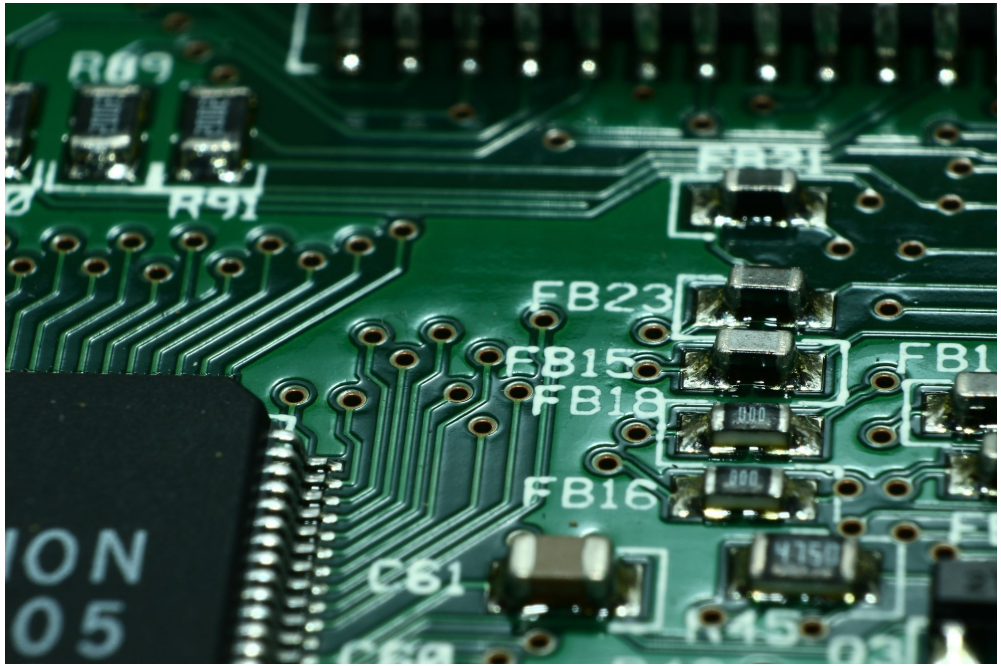
What is Soldering?

- The process of joining two metal/electrical components together
- Creates a reliable electrical connection without much special equipment



Why is Soldering Important?

- Used in everyday electronics
- Quick, durable connections
- Components can be easily built into circuit boards



SECTION II

Basics of Soldering

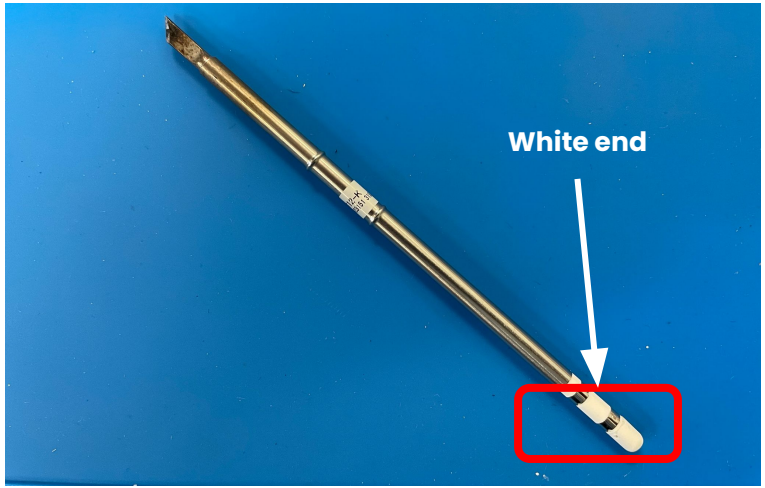
Basics of Soldering - Soldering Iron



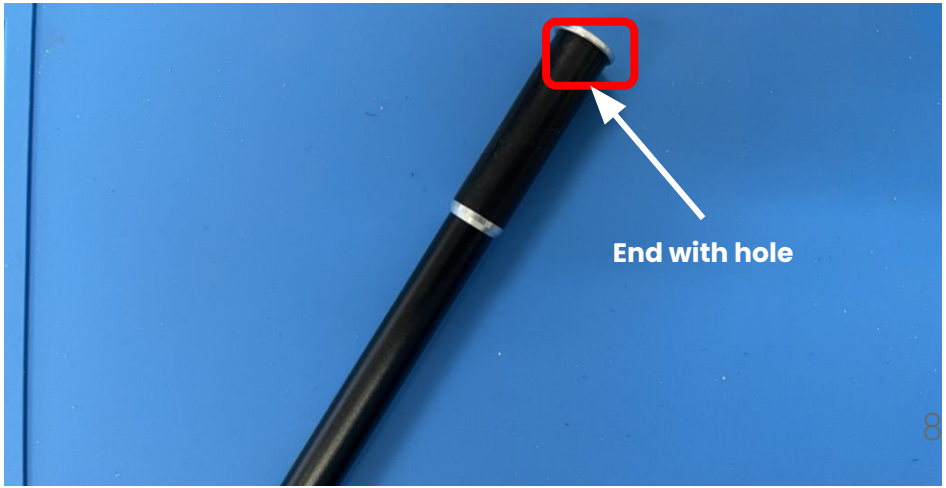
- Melts solder
- Hold like a pencil (see red rectangle)
- Do not hold the tip!
- Make sure the tip doesn’t touch anything flammable while it is on!

Basics of Soldering - Soldering Iron (continued)

- This is the iron tip
- Do not touch when heated!



- This is the iron handle
- The white end of the iron tip inserts into the hole of the iron body



Basics of Soldering - Solder

- Part that gets melted by the iron, creating the connection

Think of this less as a “glue” and more like a fusing agent that helps blend/melt the component to the board!



Basics of Soldering - Flux (Recommended)

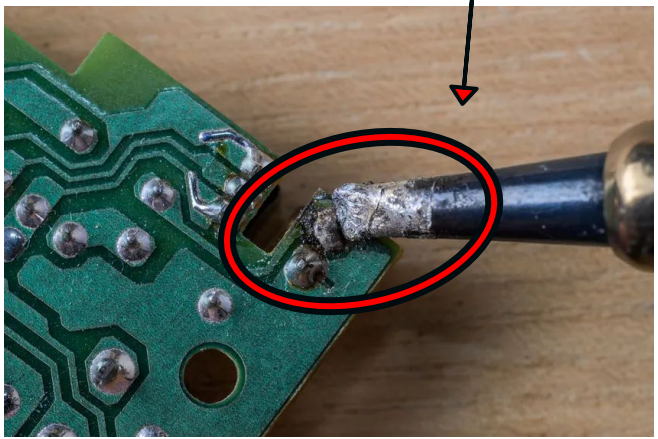
- Essentially a cleaning solution that cleans an area where solder goes
  - 90% of soldering problems are caused by oxidation
- Place on area where the solder goes on the component

Usually built into the solder, but helps if you're having trouble getting solder to stick!



See how the solder sticks more on the iron than where it's supposed to go?

USE FLUX



SECTION III

How to Solder

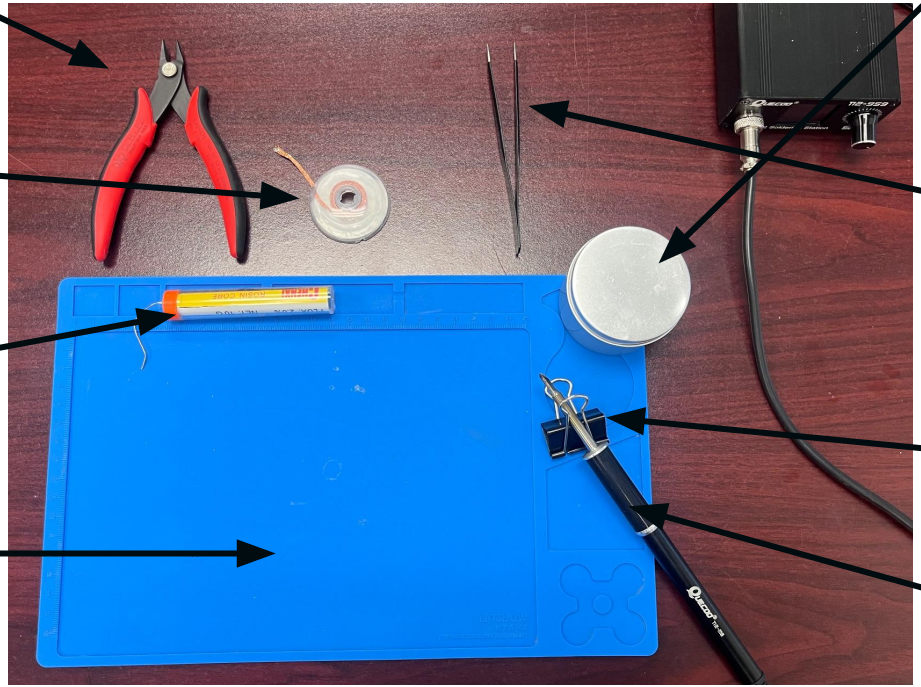
Station Layouts

Flush Cutters (Snips)
Used to cut wires

Solder Wick
Cleans up excess solder and remove components

Solder Tube
Dispenses the solder

Blue Silicone Mat
Your working area, keep all wire clippings and tools on it!



Brass Wool
Cleans tip of iron w/ rosin of oxidation and solder

Tweezers
Hold components and secure in place

Binder Clip
Holds the hot iron

Soldering Iron
Heats the solder

Safety

- Never touch iron tip directly
  - Always assume tip is hot
- Rest iron on a binder clip, not flammable objects!
- Flux and solder may contain lead
  - Don’t lick your hands
  - Don’t touch your face
  - Don’t blow on hot solder
- Components will be hot from soldering



Do not follow these images!



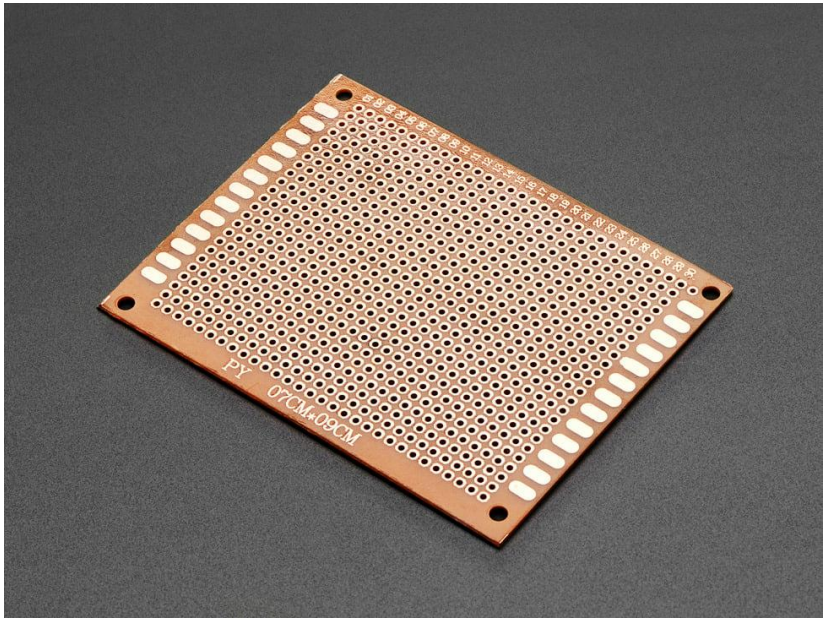
Protection

- Be sure to ventilate your area
  - **Don’t inhale fumes**
- Wear **eye protection** (safety glasses)
- Should be dressed in **standard lab uniform** (pants, shirt, closed toed shoes)
- For those with long hair, tie it back (so you don’t lose it!)
- A mask is recommended while soldering (especially without ventilation)



What is a perfboard?

Perfboard - A board with copper holes where you can stick components in and connect them together, creating solder connections.



We use a perfboard as it makes soldering and prototyping electronic circuits easier and more organized, perfect for DIY projects with fewer risks of errors.

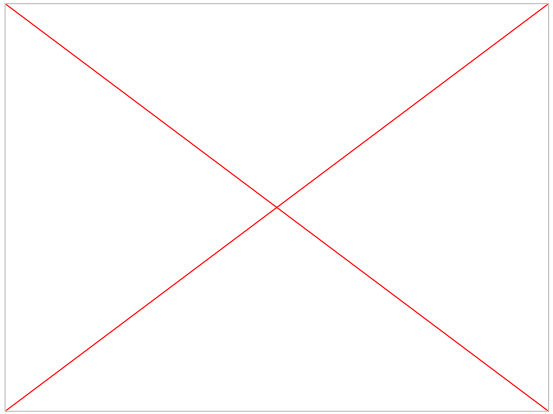
How To - Tin Tip

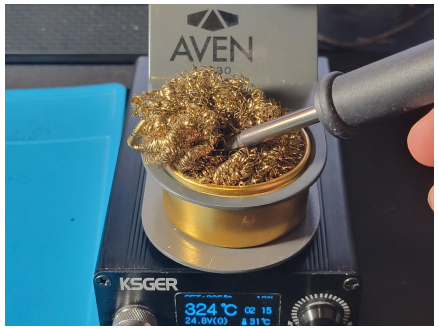
Tin the Tip to prepare and protect from Oxidation

- Heat iron to 325℃
- Apply Solder to tip
- Rotate iron tip in brass wool can to clean



(for better heat transfer)





Steps

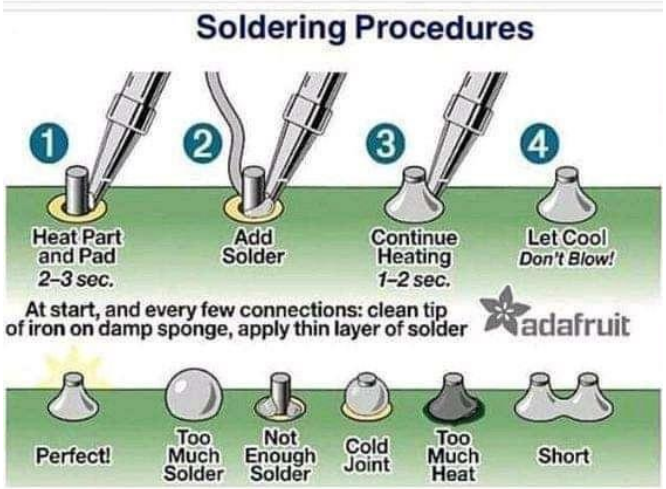
1. Prepare Board & Components
2. Assemble Components on Board
3. Confirm Circuit (Check with OPS Lab Instructor!)
4. Solder Components
5. Test Joints (with multimeter) if not working
6. Cut Leads (extra wire)
7. Clean Up Area

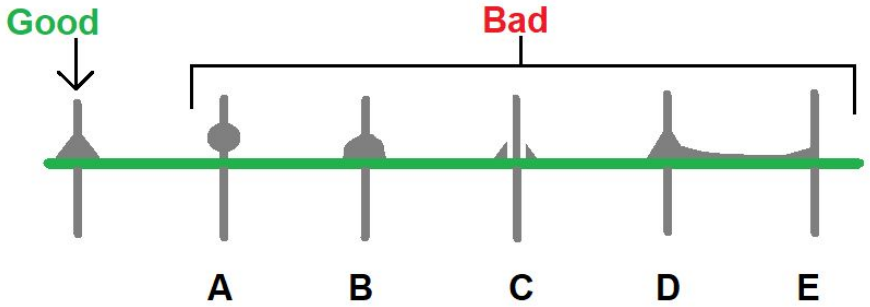
- What's wrong with this image?



Basics of Soldering - Solder Joints

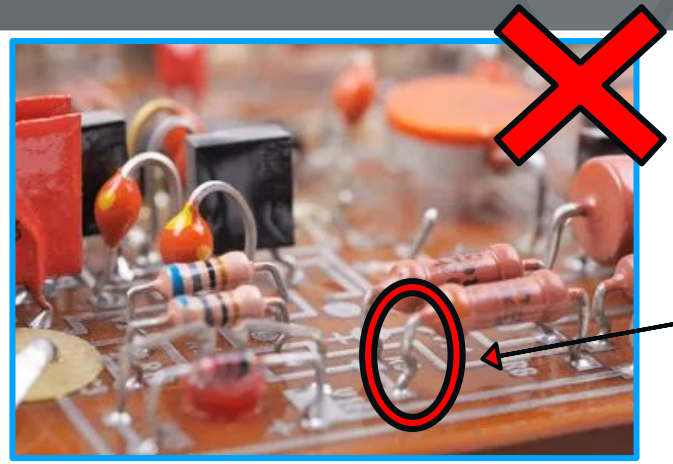
- We want OK solder joints
- Cold Joints = not enough heat to melt solder (increases resistance, could affect current)
- Insufficient Wetting, not enough rosin or too much grime
  - PCB/Components not clean
- Make sure you don't create 'unwanted bridge' between pins
- Pro tip: When dealing with resistors and components, feel free to bend the wires





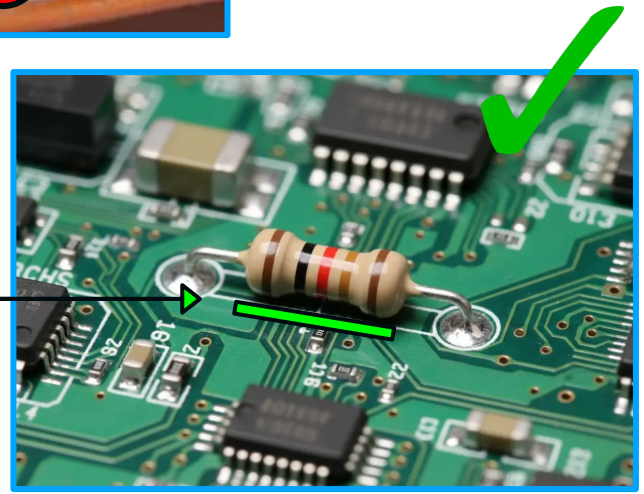
Good Component Placement

Make sure that the Component is flat and touching against the board!



Notice the space the legs create?

Notice how the main part of the resistor is flat against the board?

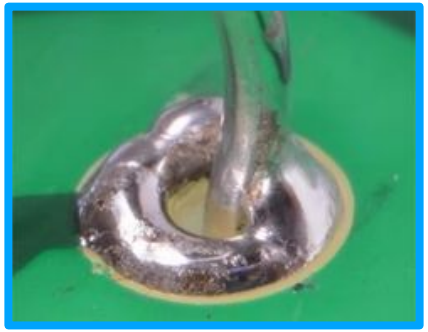


Bad solder job? You can fix it!

Have a soldering job that looks like any of these? You can fix it!



Reapply the soldering iron or wick to the blob if you have too much, let it flatten out!

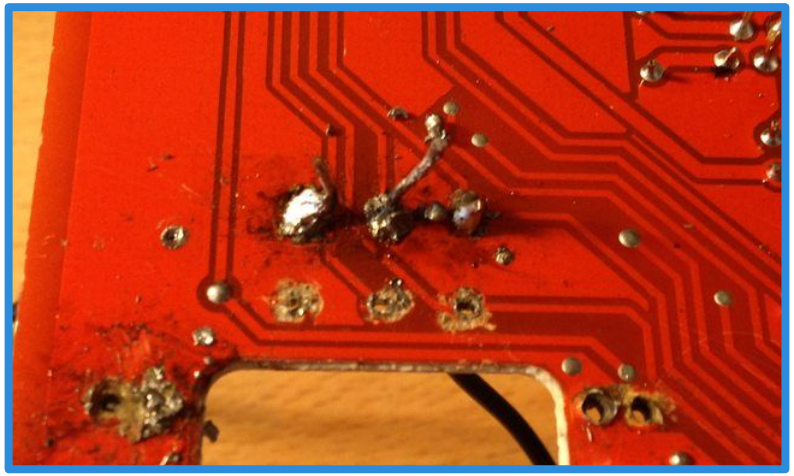


Add more solder and try again!



Heat the iron and separate the solder!

You cannot fix these :(





Avoiding touching any parts of the board too long with your iron to prevent this from happening!

Video (For perfboards)



Video #2 (For components)



Basics of Desoldering

- In the case you need to remove a connection or swap out a component, you need to **desolder**!
- Two main instruments:
  - Solder sucker (via vacuum suction)
  - Solder wick (via absorption)
- What do we need ?
  - Soldering iron
  - Brass Wool
  - Desoldering instruments (sucker or wick)

WICKED !

1. Cut a piece of the desoldering wick out
   (~3 inches for this workshop)
2. Apply a drop of flux to one end of the wick
3. Place the wick on the solder you want to remove
4. Heat up the wick and the solder using your iron
5. Keep going until the solder completely vanishes.
   (You should see the wick turn silverish from absorbing the flux)

Note: As the solder metal melts, it sticks to the wick, observe the change of the wick in color. If the solder does not stick to the wick, spread a small amount of flux on the solder and use the iron to heat it up with the wick placed on.

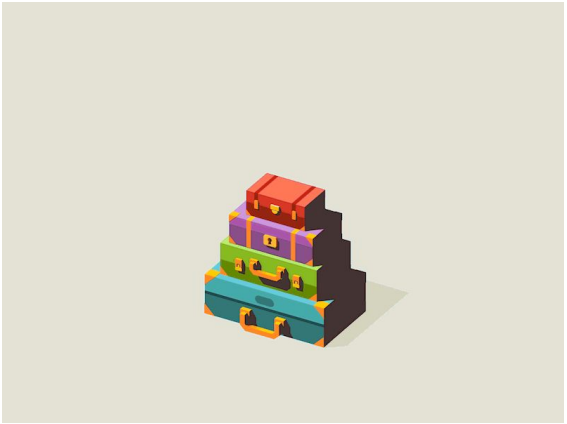
Desoldering Videos



Clean up:

1. Wipe the iron with the sponge/brass iron
2. Tin the tip (makes it easier to tin next time)
3. Turn off the iron & leave in stand to cool
4. Close brass wool tin
5. Collect trimmed leads, cooled solder and any other trash to be thrown away (and throw them away)
6. Pack up your things
7. WASH YOUR HANDS!!!!





Our Setup - The Soldering Iron

- The soldering iron comes in two parts, iron tip and iron body
- To assemble it, put the white end of the iron tip into the iron body
- To turn the iron on, toggle the switch on the back of the box the iron body is connected to
- The reading on the box's screen should be 325° C, if not, adjust with the knob on the front
- Before and after soldering, you should tin the iron
  - Tin the iron by melting a small amount of solder on the tip. If the tin does not stick to the iron, clean the iron tip using brass wool and wet sponge. If that does not work, ask a lab instructor for helps.
  - Wipe on brass wool (inside silver cylinder) to get rid of excess tin

Let the Fun Begin :))))